

Adithyaa Rettaikudi Gurumoorthi

DUAL DEGREE · B.TECH ENGINEERING DESIGN AND M.TECH ROBOTICS

CHENNAI, INDIA

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Education

Indian Institute of Technology, Madras

DUAL DEGREE - B.TECH IN ENGINEERING DESIGN, M.TECH IN ROBOTICS, CGPA: 9.05/10

Chennai, India

Nov. 2021 - May 2026

Kungliga Tekniska Hogskolan(KTH), Royal Institute of Technology

SEMESTER EXCHANGE - MASTERS IN COMPUTER SCIENCE, GPA: 9/10

Stockholm, SE

Jan. 2025 - Jun 2025

Summary

My research background spans three distinct areas: Imitation Learning frameworks, Human-Machine Interfaces (HMI), and Computational Geometry, all grounded in strong mathematical foundations. While primarily applied to robotics, my work has focus on how intelligent systems interact safely and effectively with their physical environments. I am now seeking to apply this expertise to broader, high-impact domains, including the development of responsible AI, climate research, and scientific machine learning.

Publications

INTERNATIONAL JOURNAL

1. M Reghunath, K Muralidharan, A R Gurumoorthi et al., “De (l) Noise: A Delaunay based point cloud denoising algorithm,” *Computers & Graphics*, 104310, 2025.

INTERNATIONAL CONFERENCE

1. A R Gurumoorthi, V Bagaria, B Sebastian, “Interactive steering for seamless transition during control handover in autonomous driving,” in *Proc. IEEE 4th International Conference on Smart Technologies for Power, Energy and Control (STPEC)*, 2025, pp. 1–6.

Research Experience

State-Only Offline Imitation Learning

PROF. B RAVINDRAN, PROF. NIRAV BHATT, CERA, IIT MADRAS

Aug 2025 - Jul 2026

Experimentation in Progress

- An offline imitation learning framework to enable imitator agents to learn optimal policies from sparse, **state-only expert demonstrations** (τ_e) and suboptimal, **task-agnostic imitator state-action trajectories** (τ_n)
- **Sub-Trajectory Identification**, sampling look-ahead state sequences of fixed length to capture **temporal momentum and trajectory-level intent**, avoiding the compounding errors of state transition pairs (s, s')
- Modeling **autoregressive conditional density estimators** to score expert-like sub-trajectories in unlabelled imitator datasets with the use of **contrastive likelihood-ratio scoring**
- Formulated stable **Weighted Behavioral Cloning (WBC)** objective to learn a policy from the sub-trajectories
- Benchmarked performance across continuous Gymnasium control domains; **matched performance with baselines SMODICE and LobsDICE** addressing the same task

Interactive Steering Interface for Takeover in Autonomous Driving

PROF. BIJO SEBASTIAN, AUTONOMOUS SYSTEMS LAB, IIT MADRAS

Jul 2024 - Jan 2025

Completed and Published

- Designed a physical HMI for Level 3 autonomous driving handover using a **CubeMars AK80-8 quasi-direct drive (QDD) actuator**
- Implemented a **dual-mode actuation** system: simulated a **mechanical steering dynamics** (saturated PD torque control & velocity based tire friction) in manual mode, & **position-speed control** in autonomous mode
- Engineered an intuitive control handover trigger based on **continuous resistive torque monitoring**, paired with synchronized **tri-modal feedback** (frequency-coded audio beeps, visual HUD state indicators, and haptic cues)
- Conducted **human-in-the-loop user study** in WeBots driving simulator; demonstrated a low **mean human reaction time of 0.09 s** and a **79.98% correlation** between driver performance and system satisfaction

De(l)Noise: Feature-Preserving 3D Point Cloud Denoising

Dec 2023 - Mar 2024

PROF. RAMANATHAN M, ADVANCED GEOMETRIC COMPUTATIONAL LAB, IIT MADRAS

Completed and Published

- Co-developed **De(l)Noise**, a lightweight parallel, 3D point cloud denoising algorithm that eliminates manual parameter tuning in traditional MLS and overcomes the cubic scaling bottlenecks of deep learning architectures
- Derived an **adaptive isotropic neighbourhood** via 1-ring Delaunay Triangulation (DT), incorporating circumradius filtering to reject geometric outliers and preserve local topological density
- Formulated an iterative **Moving Least Squares (MLS) hyperplane projection** with inverse-distance weighting, integrated with **PCA-based normal correction** to prevent oversmoothing across sharp edges
- The algorithm **scaled to massive point clouds** up to **20 million points** and outperformed SOTA baselines IterativePFN, PU-Net on Stanford scans, and noisy Kinect v2 datasets

Work Experience (Including Internship)

Data Scientist

Bengaluru, India

ATHER ENERGY

June 2026 - Current

- Developed **anomaly and leak detection** algorithms for EV battery packs using **multi-sensor telemetry** to enable fault identification and safety diagnostics.
- Implemented graph-based mapping pipelines to **reconstruct road geometry** and **lane networks** directly from **raw vehicle odometry**

Computer Vision and Planning Engineer Intern

Bengaluru, India

OLA KRUTRIM

May 2025 - July 2025

- Developed an autonomous driving stack for a Honda City, integrating a state-lattice **SMAC Planner** for global path search and a **Model Predictive Path Integral (MPPI)** controller for real-time dynamic trajectory tracking.
- Conducted software-in-the-loop (SIL) evaluation in **NVIDIA Isaac Sim** and **hardware-in-the-loop (HIL)** road testing on a drive-by-wire instrumented Honda City test vehicle.

Teaching Experience

[ED5340] Data Science and Theory (Lecturer: Prof. Ramanathan Muthuganapathy)

IIT Madras

TEACHING ASSISTANT

Jan 2026 - May 2026

- Created assignments and tutorial materials covering data science and theoretical concepts.
- Conducted interactive tutorial sessions to explain theoretical concepts and facilitate problem-solving discussions.

[ED2011] Design of Mechanical Systems 1 (Lecturer: Prof. Srikanth Vedantam)

IIT Madras

TEACHING ASSISTANT

Aug 2025 - Nov 2025

- Created assignments covering Hooke's Law, modulus and ultimate strength, beam bending, torsion, and buckling.
- Conducted interactive tutorial sessions to explain theoretical concepts and facilitate problem-solving discussions.

Project Experiences

Multi-Agent Systems and AI — Course Project (Advisor: Prof. Petter Ögren)

KTH

GREEDY MULTI-AGENT HYBRID A*

Feb 2025 - Mar 2025

- Developed a greedy prioritized Hybrid A* for path & motion planning in multi-agent, multi-target environments.
- Implemented a forward-backward loop for velocity profiling using path curvature and greedy predicted trajectories.
- Designed PID controller for trajectory tracking and smooth fast motion for cars and drones; placed 3rd in class.

Recent Advances in RL — Course Project (Advisor: Prof. B Ravindran)

IIT Madras

LANGUAGE-BASED HIERARCHICAL AGENT FOR SOLVING REASONING TASKS

Aug 2024 - Nov 2024

- Utilized language representation to encode image based partially observable environment for reasoning and action selection
- Built autoencoder-based feature extraction for image & language inputs to enable policy learning for downstream PPO agent

Team Abhiyaan, CFI, IIT Madras

IIT Madras

JUNIOR ELECTRONICS ENGINEER

May 2023 - Mar 2024

- Developed proprietary shared-memory bootloader & app, on ARM ISA, and testing out USB flashing of C & C++.
- Design an e-Fuse PCB for BOLT vehicle to support high systems, accommodate voltages to 48V, for IGVC competition bot.

Skills

Programming Languages

Python, C/C++, C#, Matlab, Mathematica (Wolfram)

Selected Coursework

Computational Geometry (ED5310), Nonlinear System (EE6415), Reinforcement Learning (CS6700, DA7400), Artificial Neural Networks & Deep Architectures (DD2437) (KTH), Multi-Agent Systems & AI (DD2438) (KTH), Mechanics of Serial Robots (ED6007), Linguistics (HS3017, HS5025)